Name: _____

WORKSHEET

# Two-way tables

**1** A local council wants to encourage students to ride to school. The council conducted a survey asking students if they caught a bus, rode a bike, or had other means of getting to school. Complete the two-way table below to display the results of the survey.

| | MALE | FEMALE | TOTAL |
|---|---|---|---|
| BUS | 62 | | |
| RIDE BIKE | | | 31 |
| OTHER | 14 | 23 | |
| TOTAL | | 105 | 200 |

**2** A group of students was asked if they had a dog or cat at home. Of the 30 students, 18 had a dog, and 21 did not have a cat. The number of students with a cat but no dog was 6. Use this information to complete the two-way table.

| | | DOG | | TOTAL |
|---|---|---|---|---|
| | | YES | NO | |
| CAT | YES | | | |
| | NO | | | |
| TOTAL | | | | |

**3** Two four-sided dice are thrown. Draw a two-way table to represent the sum of the two dice. Use your table to find:

**a** $P(\text{sum of } 4)$

**b** $P(\text{sum of } 1)$

**c** $P(\text{sum} > 6)$

**d** $P(\text{sum} \leq 4)$

**e** $P(\text{sum of } 7 \text{ given one die shows a } 4)$

**f** $P(\text{sum} \leq 3 \text{ given one die shows a } 1)$

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**4** Data was collected to determine if there is a relationship between the types of computers people own and gender.

| | FEMALE | MALE | TOTAL |
|---|---|---|---|
| DESKTOP | 113 | 276 | 389 |
| LAPTOP | 343 | 268 | 611 |
| TOTAL | 456 | 544 | 1000 |

**a** Rewrite the two-way table as a percentage of the total number of people and answer the following:

| | FEMALE | MALE | TOTAL |
|---|---|---|---|
| DESKTOP | | | |
| LAPTOP | | | |
| TOTAL | | | |

**i** What percentage of survey participants were female?

**ii** What percentage of people surveyed were laptop owners?

**iii** Express the male desktop owners as a percentage of the number of people surveyed.

**b** Rewrite the two-way table as a percentage of the row totals and answer the following:

| | FEMALE | MALE | TOTAL |
|---|---|---|---|
| DESKTOP | | | |
| LAPTOP | | | |
| TOTAL | | | |

**i** What percentage of survey participants were male?

**ii** What percentage of laptop owners were female?

**iii** Express the male desktop owners as a percentage of the number of desktop owners.

**iv** What can you say about gender and desktop computer ownership?

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**c** Rewrite the two-way table as a percentage of the column totals and answer the following:

| | FEMALE | MALE | TOTAL |
|---|---|---|---|
| DESKTOP | | | |
| LAPTOP | | | |
| TOTAL | | | |

**i** What percentage of survey participants were desktop owners?

**ii** What percentage of males were laptop owners?

**iii** Express the female desktop owners as a percentage of the number of females.

**iv** What can you say about females and computer ownership?

## Answers

**1**

| | MALE | FEMALE | TOTAL |
|---|---|---|---|
| BUS | 62 | 70 | 132 |
| RIDE BIKE | 19 | 12 | 31 |
| OTHER | 14 | 23 | 37 |
| TOTAL | 95 | 105 | 200 |

**2**

| | | DOG | | TOTAL |
|---|---|---|---|---|
| | | YES | NO | |
| CAT | YES | 3 | 6 | 9 |
| | NO | 15 | 6 | 21 |
| TOTAL | | 18 | 12 | 30 |

**3**

| | | DIE A | | | |
|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 |
| DIE B | 1 | 2 | 3 | 4 | 5 |
| | 2 | 3 | 4 | 5 | 6 |
| | 3 | 4 | 5 | 6 | 7 |
| | 4 | 5 | 6 | 7 | 8 |

**a** $\frac{3}{16}$ **d** $\frac{3}{8}$

**b** 0 **e** $\frac{2}{7}$

**c** $\frac{3}{16}$ **f** $\frac{3}{7}$

**4 a**

| | FEMALE | MALE | TOTAL |
|---|---|---|---|
| DESKTOP | 11.3% | 27.6% | 38.9% |
| LAPTOP | 34.3% | 26.8% | 61.1% |
| TOTAL | 45.6% | 54.4% | 100% |

**i** 45.6% **ii** 61.1% **iii** 27.6%

**b**

| | FEMALE | MALE | TOTAL |
|---|---|---|---|
| DESKTOP | ≈29.05% | ≈70.95% | 100% |
| LAPTOP | ≈56.14% | ≈43.86% | 100% |
| TOTAL | 45.6% | 54.4% | 100% |

**i** 54.4% **ii** 56.14% **iii** 70.95%

**iv** Based on the given data, desktop computer owners are more than twice as likely to be male than female

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**c**

| | FEMALE | MALE | TOTAL |
|---|---|---|---|
| **DESKTOP** | ≈24.78% | ≈50.74% | 38.9% |
| **LAPTOP** | ≈75.22% | ≈49.26% | 61.1% |
| **TOTAL** | 100% | 100% | 100% |

**i** 38.9% **ii** 49.26% **iii** 24.78%

**iv** Based on the given data, about 3 out of 4 females own a laptop while 1 in 4 own a desktop.